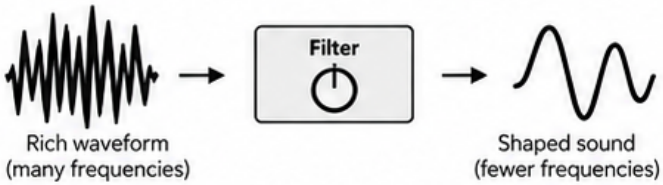
Types of Music Synthesis

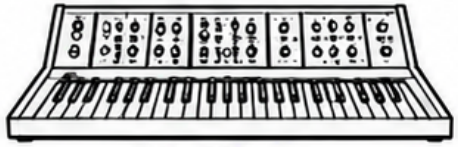
Different ways to create sound using technology.

Subtractive Synthesis

Start with a rich sound (lots of frequencies) and remove parts using filters, envelopes and modulation.



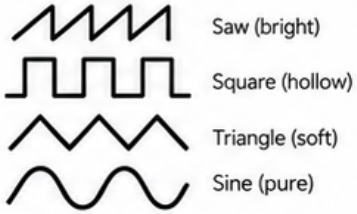
Common sounds: classic synth sounds, basses, pads, leads.
Examples: Moog Minimoog, Roland Jupiter-8, Korg MS-20, Roland SH-101.



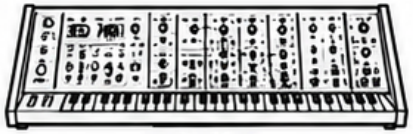
Moog Minimoog

Analog Synthesis

Uses electronic circuits to generate basic waveforms (saw, square, triangle, sine) and shapes them with filters, envelopes and modulation.



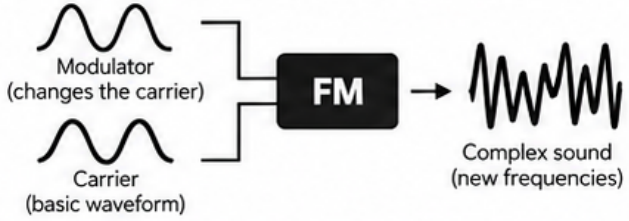
Common sounds: warm pads, basses, leads, rich evolving sounds.
Examples: Moog Minimoog, Roland Jupiter-8, Korg MS-20, ARP 2600.



ARP 2660

FM Synthesis

One waveform (the modulator) changes the shape of another waveform (the carrier), creating new and complex sounds.



Common sounds: bells, electric pianos, metallic and evolving sounds.
Examples: Yamaha DX7, Yamaha TX81Z, Korg DS-8, Elektron Digitone II.



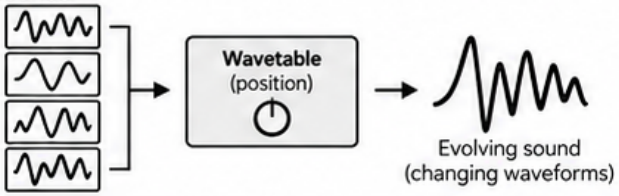
Yamaha DX7



Elektron Digitone II

Wavetable Synthesis

Uses a table of many waveforms and moves through them (or blends between them) to create changing sounds.



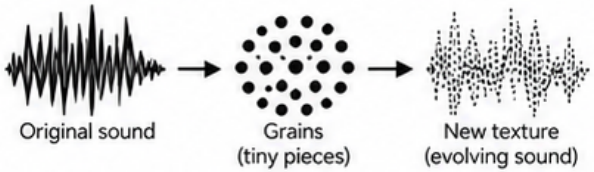
Common sounds: dynamic pads, evolving textures, digital sounds.
Examples: Korg Modwave MK2, PPG Wave, Waldorf Microwave.



Korg Modwave MK2

Granular Synthesis

Breaks a sound into tiny pieces (grains) and reassembles them in different ways to create new textures and evolving sounds.



Common sounds: atmospheres, textures, soundscapes, evolving pads.
Examples: Tasty Chips GR Mega, Granulator II, Absynth, Quanta.



Tasty Chips GR Mega

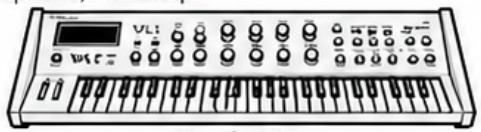
Physical Modelling

Uses maths to simulate real instruments (e.g. strings, brass, woodwinds) rather than recording them.



Model of a vibrating string
(changes over time)

Common sounds: realistic acoustic instrument sounds, strings, brass, woodwinds, plucked sounds.
Examples: Yamaha VL1, Applied Acoustics Systems Chromaphone, Pianoteq.



Yamaha VL1